

Temma Driver

Table of contents

Introduction	3
Revision History	3
Getting Started	3
System requirements	3
Installing	4
Using the Temma Driver with Client Apps	4
Getting help and reporting driver issues	5
Temma Mount Setup Window	6
Initialization	7
Mount Configuration	8
Site Setup	9
Telescope	9
Temma Setup Dialog	9
Guide Correction Speeds	9
High Precision Slew	10
Parking	10
Power Voltage	11
Syn to Alt/Az	11
Sync To Zenith	11
Tracking Rates	11
Troubleshooting	13
Resources	13
Cable Drawings	14
Driver Updates	15

Introduction

This help file describes how to configure and operate the Temma V4 ASCOM Telescope driver.

The Temma V4 telescope driver supports the ASCOM Telescope Driver Interface standard V4.

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Revision History

1.0.0 - Initial release ([see Github](#))

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Getting Started

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System requirements

Software

- 1.A computer running in Windows 10/11 (32-bit or 64-bit).
- 2.ASCOM Platform 6 (or later) must be installed.
- 3..Target Net Framework 4.7.2.

Hardware

1. Takahashi Temma 2, or Temma 2M mount.
2. A computer with a serial port or USB/Ethernet to serial port converter.

Note: Users should avoid Keyspan USA-19HS serial adapters, BSOD crashes occur when used with the ASCOM Temma driver. The crash dump file points to the Keyspan driver, this issue was reported to Tripp Lite on 12/12/10. BSOD occur from flaws in kernel level drivers, the ASCOM TEMMA driver is user level, indicating a hardware fault or a buggy kernel mode Keyspan driver.

3. A Serial cable to connect a PC to the telescope.
4. Power Source

Select the appropriate power setting from the 'Power' drop-down box. You must have the appropriate 24 volt power supply to run at the 24 volt high slew rates.

High Speed Drive	Working Voltage	
	DC12V	DC24V

EM-400 Temma2	250X	500X
EM-10 Temma2 Jr.	120X	250X
EM-200 Temma2 Jr.	120X	250X
EM-200 Temma2	350X	700X
NJP Temma2	175X	350X
EM-500 Temma2	260X	520X
EM-200M	700X	Not Supported
EM-400M	500X	Not Supported
EM-12M	150X	Not Supported

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Installing

Before installing the Temma V4 telescope driver please make sure you have installed the ASCOM 6.0 (or later) platform. You can download it from <http://www.ascom-standards.org>. You will also need to install the Microsoft .Net Framework 4.7.2 and possibly service packs. More details on the requirements can be found on the ASCOM-Standards site.

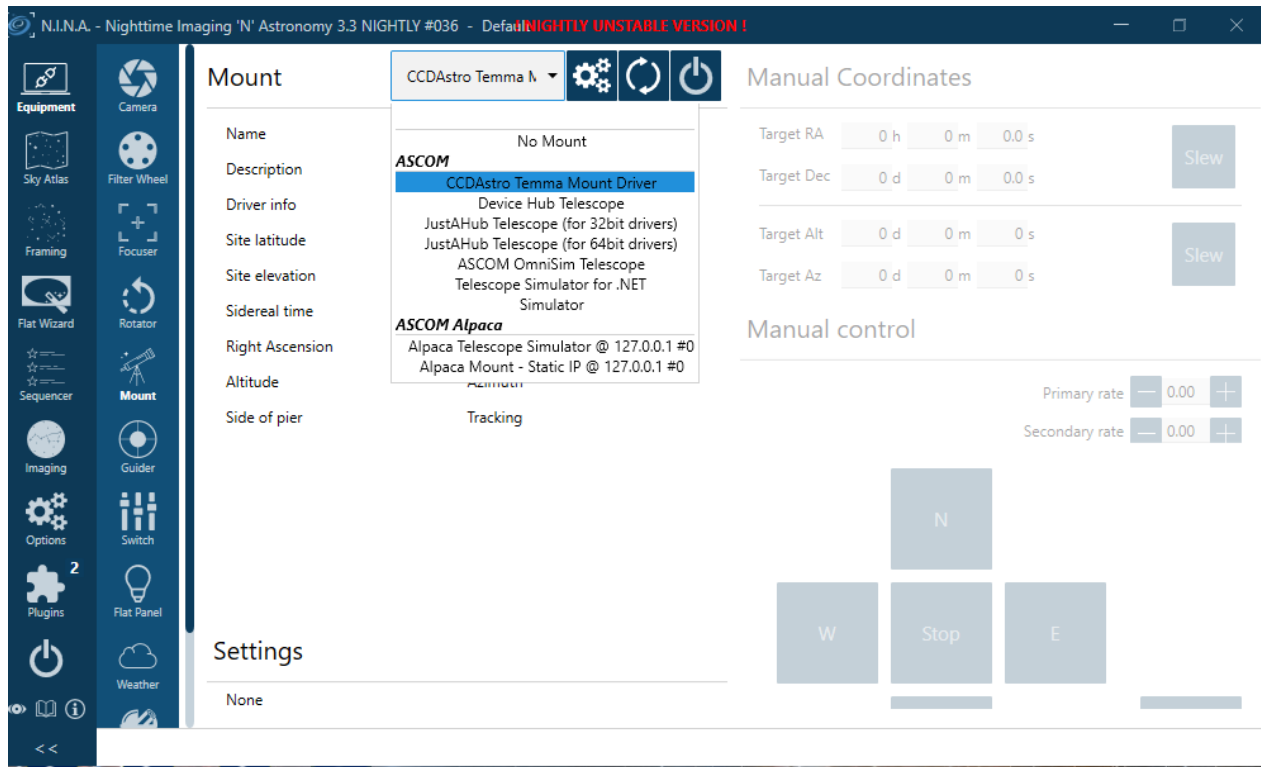
You can always install any version of the V4 driver without uninstalling the previously installed version of the V4 driver.

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Using the Temma Driver with Client Apps

To use the driver select it in an ASCOM application.

For example, in NINA click the Telescope then choose 'CCDASTRO Temma Mount Driver'



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Getting help and reporting driver issues

In order of preference potential driver problems can be reported via several methods:

E-mail to ccdastroinc@gmail.com.

IMPORTANT: When reporting a potential ASCOM driver bug here are some general guidelines:

1. The first aim of a bug report is to let us see the failure with our own eyes. Please give us detailed instructions so that we can make it fail for ourselves.
2. The second aim of a bug report is to describe what went wrong. Describe everything in detail. State what you saw, and also state what you expected to see. Write down the error messages, especially if they have numbers in them. Better yet, take a screenshot!
3. By all means try to diagnose the fault yourself if you think you can, but if you do, you should still

report the symptoms as well.

4. Be ready to provide extra information if we need it, like trace logs.
5. If other programs are involved state the EXACT version number of each program. Usually you can find this information in the programs Help->About menu item.
6. Write clearly. Say what you mean, and make sure it can't be misinterpreted!

Thank you in advance for following these guidelines! For more details please see this link:

[Github](#)

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Temma Mount Setup Window

ASCOM Setup

Telescope Setup window:

Detailed explanations of each field follows in the next sections.

Temma Setup

Mount Setup

Com Port: COM1

Tracking Rate: Sidereal

Voltage: 24V

Model: EM200

☒ Enable Trace Logging

Site Information Setup

Latitude: N 27 0 0

Longitude: W 80 0 0

El (m): 15

Telescope Orientation

☐ OTA East - Scope Pointing West

☐ OTA West - Scope Pointing East

☒ Counterweight Down ☒ Keep Last Sync

☐ Counterweight West ☐ Ask At Start

Telescope Setup

Telescope Aperture (mm): 0

Central Obstruction (mm): 0

Focal Length (mm): 0

☐ Hi-Precision GOTO

☐ Tracking Off on Connection

☐ Warm Before Meridian Flip

Park/Rate Settings

Park Settings

0 Alt - Az 0

☒ Current Position ☐ Slew To

☐ Unpark On Reconnect

Guide Settings

0.50 Ra - Dec 0.50

☐ Send Rate

OK Cancel

Initialization

Temma Setup and Initialization Procedure

1. Turn off the Temma telescope mount.
2. Connect the RS232 serial port on the Temma to an open serial port on your computer.
3. Turn the Temma Power switch on. Each time the Temma is powered on, the Temma control system is initialized with the following default parameter:
 - * The optical tube assembly (or telescope) is assumed to be on the West side of the mount.
 - * The telescope is pointing to 0.00 in both right ascension and declination.
 - * Right ascension and declination correction speeds are set to 90 (unless you have entered a different rate in the Temma driver's main window).
 - * The control system tracking is for the Northern Hemispheret.
4. Open your planetarium software.
5. In the 'ASCOM Telescope Chooser Control list, select **Temma by Takahashi**.
6. Click **Settings** and then select the correct COM port and input your info in the other fields (e.g.: longitude/latitude).
7. Establish a link to the Temma.
8. Select one of the orientation modes from the 'Init' windows
9. When the link is established, center the appropriate star as directed and then synchronize the planetarium software (e.g.: for TheSky press its 'Sync' button).

Synchronizing

Procedure

To synchronize with the optical tube assembly on the East side of the mount, Scope pointing West

1. Loosen the clutches on the mount and move the telescope to the East side of the mount.
2. Center a known star that is on the West side of the Meridian in the field of view.
3. Identify the star in your planetarium software.
4. Sync on the star.
5. An on screen dialog will notify you of the correct side of the meridian to select the sync star.

To synchronize with the optical tube assembly on the

1. Loosen the clutches on the mount and move the telescope to the West side of the mount.

- | | |
|--|---|
| West side of the mount,
Scope pointing East | <ol style="list-style-type: none"> 2. Center a known star that is on the <i>East</i> side of the Meridian in the field of view. 3. identify the star in your planetarium software. 4. Sync on the star. 5. An on screen dialog will notify you of the correct side of the meridian to select the sync star. |
|--|---|

**Scope Pointed at Pole,
Counter-weight down
position**

This method is intended for remote users. The CW down method gives a rough initial alignment that must be refined by a subsequent sync for accurate GOTO operations. It is recommended that the subsequent sync be accomplished using ACP (Pinpoint) to plate solve a CCD image after the initial alignment. Obviously, a manual/visual sync can be used if desired.

**Scope Pointed at Pole,
Counter-weight West**

The CW West method gives a rough initial alignment that must be refined by a subsequent sync for accurate GOTO operations. Place the CW on the West side of mount with the CW shaft and OTA parallel to the ground.

Keep Last Sync

The Temma telescope was previously synced and not powered off, or the Temma was 'Parked'

SyncZ

New feature: 'Sync Z' allows the user to initialize and accurately sync the mount in the Takahashi 'pointed at the Zenith' position. To use this feature first turn off tracking, point the scope at the Zenith (using a level helps position the scope accurately), then click the 'Sync Z' button.

- * power on the mount
- * start the driver and turn off tracking
- * polar align
- * move the scope manually to point to Zenith (use a level for best results)
- * press SyncZ and the mount starts with RA=sidereal time and Dec=Latitude

Mount Configuration

Mount Type: Select the mount type that matches your mount from the drop-down box.

Site Setup

Site Setup

Latitude: Latitude of the selected site. Latitude must be in degrees: minutes: seconds format.

Longitude: Longitude of the selected site. Longitude must be in degrees: minutes: seconds format.

Elevation: Elevation in meters

To get Longitude/Latitude from TheSky: clicking the button will retrieve the longitude and latitude from Thesky (check the 'X' box if you are using Thesky X). The Longitude/Latitude fields will be populated with the data from Thesky's site data .

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Telescope

Aperture: Enter the primary telescope's aperture in millimeters.

Focal Length: Enter the primary telescope's focal length in millimeters.

Note: these fields are not specifically used by the driver but they are passed up to clients if the client asks for them.

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Temma Setup Dialog

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Guide Correction Speeds

Guide Rates

Set

Enter the right ascension and declination correction speed (as a percentage of the Sidereal rate, 10 - 99%). This setting is used for the slow speed moves or for guiding.

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High Precision Slew

High precision slew

Check the 'Hi-P Slew' checkbox (marked by red arrow below) to activate this feature to reduce the Temma's inherited encoder resolution errors after a GOTO slew. This offers an automated centering that may be faster and better than doing manual corrections to bring objects to the of the FOV.

Special thanks to Hartmut Bornemann for supplying the code snippets and support implementing this new feature.

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Parking

Parking the Temma

The Temma's electronics do not have the ability to park the mount via its firmware. However, this driver offers a solution via software.

Park Procedure:

Two park methods are available:

'Slew Before Park' slews to the inputted alt/az position before parking
 'Park at Current Position' parks the current alt/az position of the mount.

	Prior to using the park feature the mount must be correctly synced as described above.
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	The 'Set Park' button can be used to select the mount's current position as the parking position whenever the Temma is connected. The position will be saved thus removing the need to manually input the values.
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1. Input the desired park altitude and azimuth
2. Select one of the park methods
3. Click the 'Park Mount' button
4. When the park slew is completed disconnect the driver and immediately power off the mount or the mount can be left parked.

However, if the power is not turned off the mount will remain in the simulated 'tracking off' mode for NON-Temma 2 models.



Upon reconnection to the mount the unpark sequence will begin.

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Power Voltage

Displays the mount's voltage.

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Syn to Alt/Az

The 'Sync to Alt/Az' and the 'Slew Prior' button is used to re-sync the mount to the Altitude and Azimuth shown in the user input boxes. This feature is useful when the mount appears to be lost after an unpark.

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Sync To Zenith

'Sync Z' allows the user to initialize and accurately sync the mount in the Takahashi 'pointed at the Zenith' position.

To use this feature:

- * power on the mount
- * start the driver and turn off tracking
- * polar align
- * move the scope manually to point to Zenith (use a level for best results)
- * press SyncZ and the mount starts with RA=sidereal time and Dec=Latitude

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Tracking Rates

Tracking Rates

- ☐ Sidereal (default)
- ☐ Lunar - When used in conjunction with the 'TheSky Format' checkbox the lunar rate will be set to a high precision mode that will control both the RA speed and Dec rate to account for parallax
- ☐ Solar
- ☐ King - Rate will compensate for atmospheric refraction
- ☐ Manual - When selected rates must be inputted into the RA and DEC boxes, then press the 'Send' button
- ☐ Custom - This feature requires TheSky v6. When selected users can input the object's name and update interval in to the 'Custom Tracking Rates' frame. Updates to current offsets will be made at the desired intervals.

Clicking on the 'Tracking Rate' option buttons will toggle the rates.

'Tracking On/Off' is used to stop/start the tracking on Temma 2 mounts, or slow the tracking rate to ~60% sidereal rate for **non** Temma 2 mounts. This is useful if you want to pause viewing. The slower rate will avoid the mount running in to itself when left unattended. Checking the 'Rewind RA' box in the Misc frame (when using **non** Temma 2 mounts) will rewind the right ascension back to the point where the tracking was stopped adding one more level of mount protection for unattended pauses in observing or imaging.

Custom/Comet Tracking Rates

RA Adjustment (sidereal sec/day)

- ☐ Enter the adjusted tracking rate for the right ascension axis in *sidereal* seconds per day max = +/- 32767. There are 86,164 seconds in one *sidereal* day (1 "earth day" = 1.002738 "sidereal day").

Dec Adjustment (min/day)

- ☐ Enter the adjusted tracking rate for the declination axis in minutes per day, max = +/- 1180 (plus = North, - = South).

Example:

1. +140, +50 - Entering **+140** into the **RA** text box will speed up the RA tracking by 140 seconds per day (one revolution normally takes 86164 seconds at sidereal rate), +50 into the **Dec** text box will move the declination axis 50 minutes per day North. Check the 'Manual Input' box 'Send' to apply the new comet settings. The maximum offsets are +/- 32767 for RA, and +/-1180 for Dec

TheSky Format

- ☐ **Checking this box allows for the direct input of RA/Dec rates obtained from TheSky's object info box**

- ☐ Input the name of the object (as recognized by Thesky) and the data will be inputted and used automatically to calculate custom rates when the 'Use Thesky' box is checked.

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Troubleshooting

Troubleshooting:

The telescope crosshair in my planetarium software jumps	The coordinates reported by the Temma control system (when the telescope is not slewing) can vary by one arcsecond. This causes the telescope cross hairs to "jump" between the reported positions. The Temma protocol reports coordinates to the nearest second in right ascension (15 arcseconds at the 0 degrees declination) and to the nearest 6 arcseconds in declination. The Temma's control system accepts declination coordinates to the nearest 12 arcseconds, so, at small fields of view, the cross hairs will not exactly match the coordinate that was used to synchronize the telescope These are limitations of the Temma protocol.
The telescope crosshair in TheSkyX does not move when slewing or syncing	Increase the crosshair update frequency to 1000 ms or more.

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Resources

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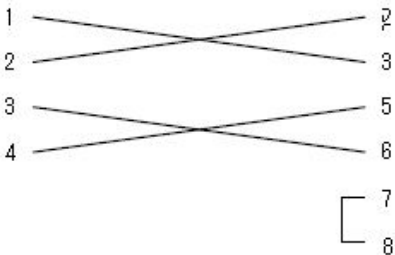
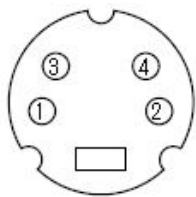
Cable Drawings

----- RS-232C CABLE -----

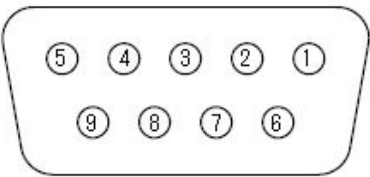
for

- Temma, TemmaPC, Temma2
- TemmaJr, TemmaPCJr, Temma2Jr

MINI DIN 4P MALE
(TOP VIEW)

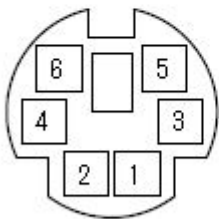


D-sub 9P FEMALE
(TOP VIEW)



----- AUTO GUIDE CONNECTOR -----

- USD, USD II, USD3
- Temma, TemmaPC, Temma2
- TemmaJr, TemmaPCJr, Temma2Jr
- PD-7XY, PD-8XY



MINI DIN 6P FEMALE
(TOP VIEW)

No.	
1	NC
2	RA(+)
3	RA(-)
4	DEC(↑)
5	DEC(↓)
6	GND

Driver Updates

Get the most current driver [here](#).

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